

Databases 2 - exam - June 17, 2024 - Dur. 2h

S. Comai, P. Fraternali, D. Martinenghi

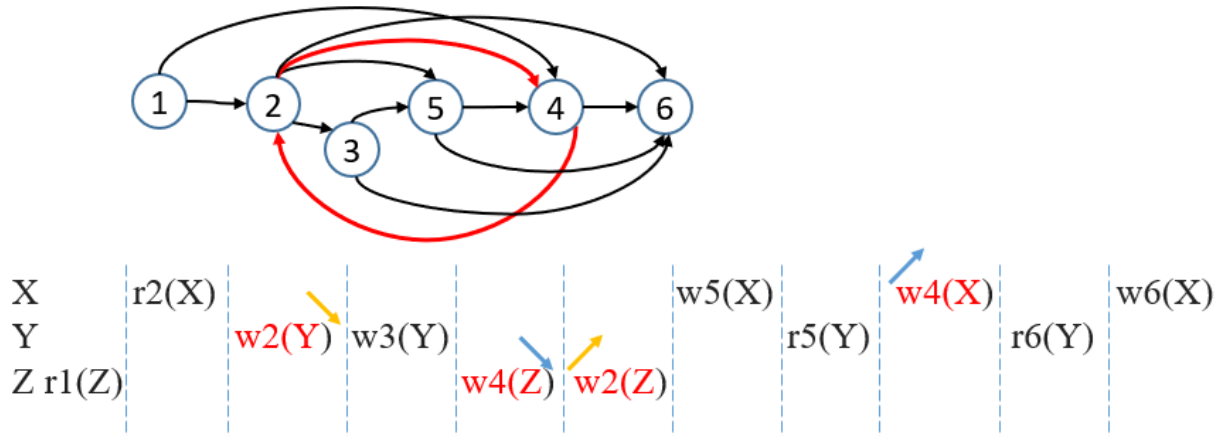
A. Concurrency control (10 points, up to 2 per class)

Classify the following schedule with respect to VSR, CSR, 2PL, Strict 2PL, and TS multi (with the conventions adopted for TS Multi under Snapshot Isolation, used for the exercises). Motivate all your answers.

If it is VSR, provide all the possible serializations. If it is not 2PL, explain which lock/unlock requests cause this.

r1(Z) r2(X) w2(Y) w3(Y) w4(Z) w2(Z) w5(X) r5(Y) w4(X) r6(Y) w6(X)

Solution.



- CSR: NO - there is a cycle due to r2(X) w4(X) and w4(Z) w2(Z)
- VSR: NO – swapping the blind writes w5(X) and w4(X) does not remove the cycle
- 2PL: NO - w2(Z) cannot anticipate the lock request and w2(Y) must release the lock before w3(Y)
- w4(X) cannot anticipate the lock before w5(X) and w4(Z) must release the lock before r2(Z) - to determine the belonging to 2PL, it would also suffice to notice that it is not in CSR, so it cannot be in 2PL
- Strict 2PL: NO, it is not 2PL
- TS multi: NO, w2(Z) cannot be after w4(Z)

B. Ranking (12+2 points, up to 3 for Questions 3 and 5, up to 2 for the remaining questions)

A group of people from Milan and Como are looking for a meeting point allowing them to easily travel back and forth. They want to minimize the *mean* travel time m to both cities and to have balanced times. Thus, they devise the following scoring function for any place with a travel time x to Milan and y to Como:

$$f(x, y) = m + s, \text{ with } m = \frac{x + y}{2} \text{ and } s = \sqrt{(x - m)^2 + (y - m)^2},$$

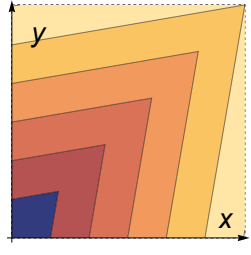
where m is the mean time and s is akin to the standard deviation of the population $\{x, y\}$ (geometrically, s is the distance of $\langle x, y \rangle$ to the bisector of the first quadrant of the Cartesian plane, where $x = y$, i.e., times are balanced). The group is only given three options, A , B , and C , vertically distributed over two rankings as follows (for your calculations, consider that $f(20, 100) \approx 116.57$):

x	y
B : 20	A : 20
A : 100	B : 100
C : 110	C : 110

1. What is the skyline of this very small dataset? Do not show any step or calculation, just the result.
2. Compute the top-1 meeting point with TA according to $f(x, y)$. Show depth and accesses.
3. The result found by TA is not a skyline point. Explain how this is possible.
4. Compute the top-1 meeting point with FA according to $f(x, y)$. Show depth and accesses.
5. The result found by FA is wrong. Explain how this is possible.
6. (**Bonus** for the more mathematically inclined) Can you draw the shape of the iso-score curves of f ?

Solution.

1. The skyline is $\{A, B\}$.
2. A and B are discovered during round 1, with the same score ≈ 116.57 (they are symmetric and so is f). The threshold is $T = 20$ at round 1 and $T = 100$ at round 2 (the threshold point lies on the bisector, so the distance to it is 0). At round 3, C is found, with $f(C) = 110$ (C is on the bisector), and $T = 110$. We stop and output C . Depth=3, 6 s.a., and 3 r.a. (or 2, depending on implementation).
3. The skyline is the set of all optimal points according to some *monotone* function, but f is not monotone.
4. FA stops at depth 2 (with 4 sorted accesses), discovering only A and B and makes no random access in this case. A and B are tied at 116.57; either is returned.
5. Again, f is not a monotone scoring function and FA's correctness requires a monotone function.



6. A few iso-score curves are shown here:
Indeed, the locus of points having a score c is characterized as follows:

$$f(x, y) = c \Leftrightarrow \text{(by substitution)} \frac{x + y}{2} + \sqrt{\left(\frac{x - y}{2}\right)^2 + \left(\frac{y - x}{2}\right)^2} = c \Leftrightarrow x + y + \sqrt{2}|x - y| = 2c$$

When $x > y$ this reduces to the straight line $y = x \frac{\sqrt{2}-1}{\sqrt{2}+1} - \frac{2c}{\sqrt{2}-1}$, and for $x < y$ to $y = x \frac{\sqrt{2}-1}{\sqrt{2}+1} + \frac{2c}{\sqrt{2}+1}$ (and when $x = y$ then $x = y = c$).

C. Physical databases (10 points)

Table **Customer**(CUSTID, LastName, FirstName, Country) contains 800K tuples (primary key is in capital letters) and **Purchase** (PRODUCTID, CUSTID, DATE, Qty) contains 8M tuples. Estimate the cost of running the following query in the scenarios described below. Describe an efficient query plan for the following scenarios. Estimate their execution costs (**write the complete formula for the query**) and explain all the steps of the plan and their costs. The evaluation of the exercise also considers the plan's degree of efficiency.

```
SELECT *
FROM Customer C join Purchase P ON C.CUSTID=P.CUSTID
WHERE Country=\Italy" or Country=\Spain"
```

1. Table **Purchase** is entry-sequenced and occupies 40K block. Table **Customer** is stored in a hash table with a hash function defined on the key. It occupies 8K blocks and has a negligible overflow chain. Val(Country)=160.
2. Like in point 1 plus: Table **Purchase** has a secondary hash index built on attribute **CustID** with the same hash function of **Customer** but with a cost of access due to overflow of 1.3. With the overflow blocks, it occupies 10.5K blocks.
3. Like in point 2 plus: Table **Customer** has a B+ index (3 levels, 1K leaf nodes) with search key **Country**.

Solution.

1. **Plan 1** (first scan **Customer**): Scan (sequentially) table **Customer**. For each tuple satisfying Country="Italy" or Country="Spain", scan table **Purchase** and compute the join condition.

$$\text{Plan1} = 8K + 2 \cdot (800K/160) \cdot 40K = 8K + 10K \cdot 40K = 400M \text{ I/O accesses}$$

Plan 2 (first scan **Purchase**): Scan table **Purchase**; for each customer access the hash table and check if Country="Italy" or Country="Spain".

$$\text{Plan2} = 40K + 8M \cdot 1 = 8,04 \text{ M I/O accesses (best plan)}$$

Plan 3 (Nested loop join):

$$\text{Plan3} = 8K + 8K \cdot 40K = 320 \text{ M I/O accesses}$$

2. **Plan 4** (improve Plan 1): Scan (sequentially) table **Customer**; for each tuple satisfying Country="Italy" or Country="Spain", find **CustID** in the hash index and follow the pointers to retrieve the purchases of the customer.

$$\text{Plan4} = 8K + 2 \cdot (800K/160) \cdot (1.3 + 10) = 8K + 10K \cdot 11.3 = 121K \text{ I/O accesses}$$

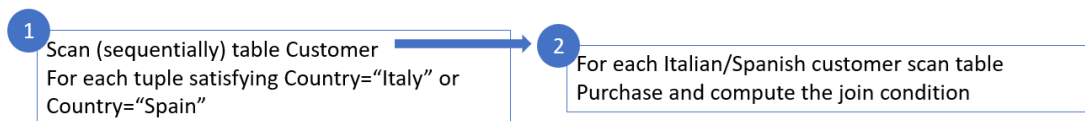
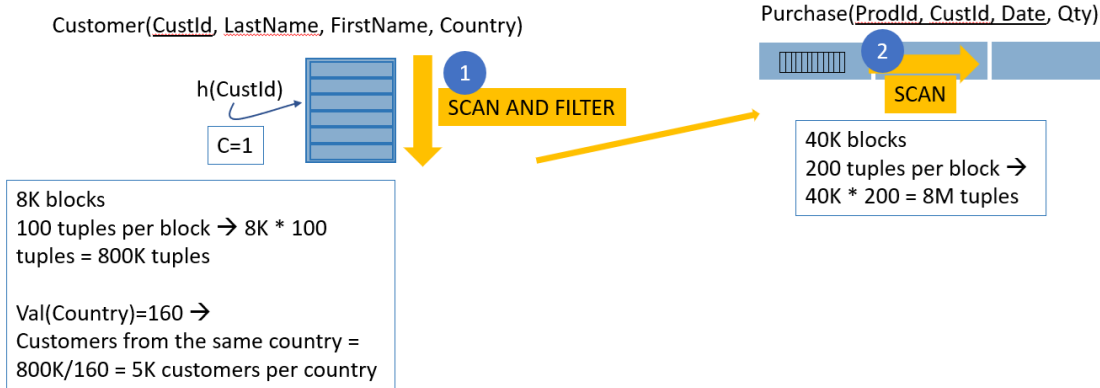
Plan 5 (Hash join + retrieve purchases): compute the hash join; for each Italian/Spanish customer, retrieve their purchases.

$$\text{Plan5} = 8K + 10.5K + 2 \cdot (800K/160) \cdot (8M/800K) = 8K + 10.5K + 10K \cdot 10 = 118,5K \text{ I/O accesses (best plan)}$$

3. **Plan 6** (extend Plan 4): Find Italy and Spain in the B+ index, follow the pointers to retrieve Italian and Spanish customers, then lookup customers in the hash index and retrieve their purchases.

$$\text{Plan6} = 2 \cdot (2 + (5K/800) + 5K) + 10K \cdot (1.3 + 10) = 123K \text{ I/O accesses (it does not improve Plan 4!)}$$

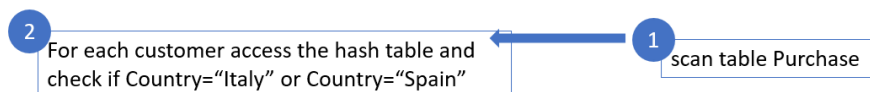
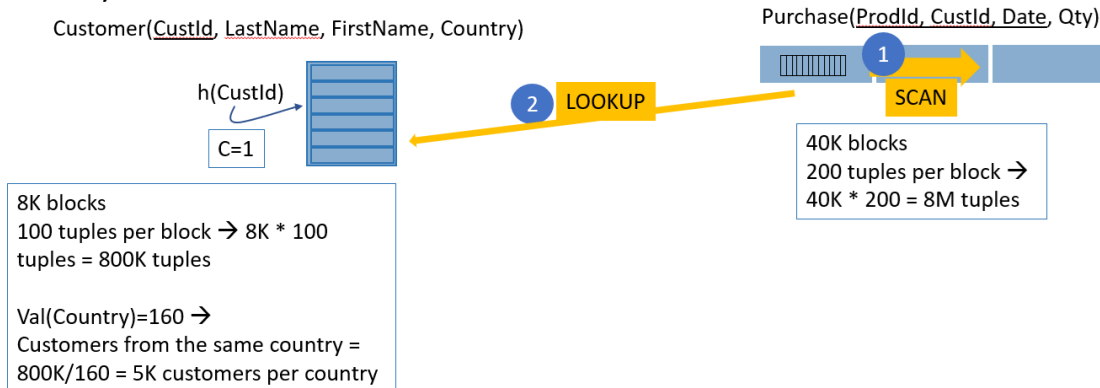
SCENARIO 1) – first scan CUSTOMER



$$\text{Plan1} = 8K + 2 * (800K/160) * 40K = 8K + 10K * 40K = 400M \text{ I/O accesses}$$

Scan Customer + for each filtered tuple scan Purchase

SCENARIO 1) – first scan PURCHASE



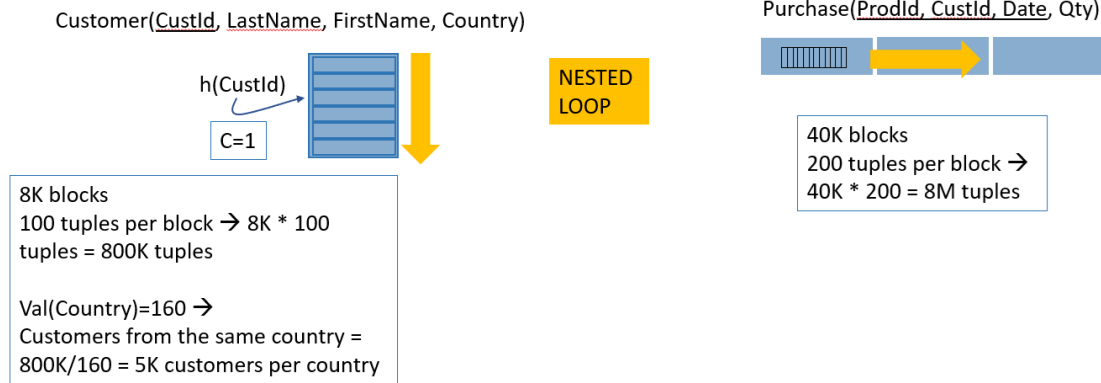
$$\text{Plan2} = 40K + 8M * 1 = 8,04 M \text{ I/O accesses}$$

Scan Purchase + for each purchase search CustID and check condition



Best plan for scenario 1)

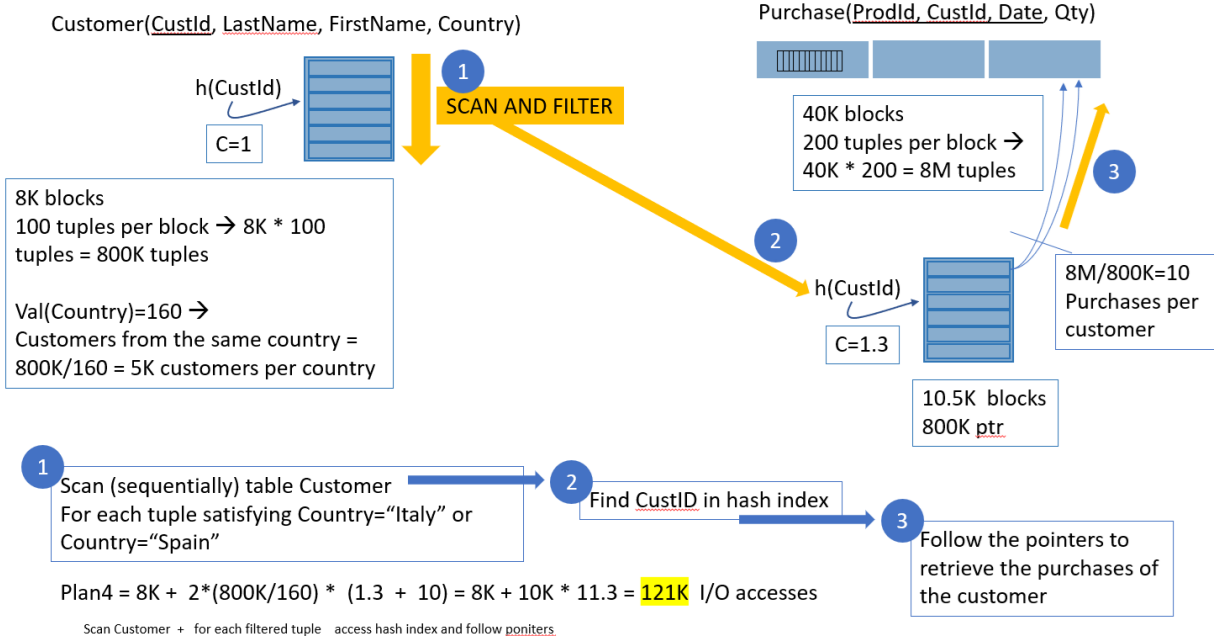
SCENARIO 1) - NESTED LOOP JOIN



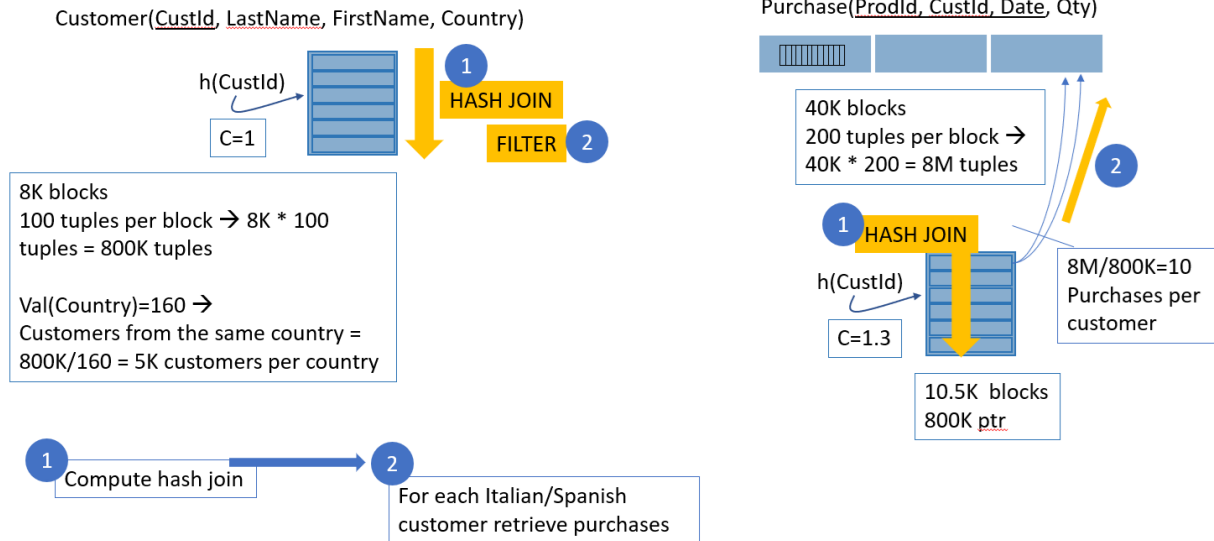
$$\text{Plan3} = 8K + 8K * 40K = 320 \text{ M I/O accesses}$$

Scan Customer + for each block scan Purchase

SCENARIO 2) – Improve Plan 1



SCENARIO 2) – HASH JOIN + Retrieve purchases



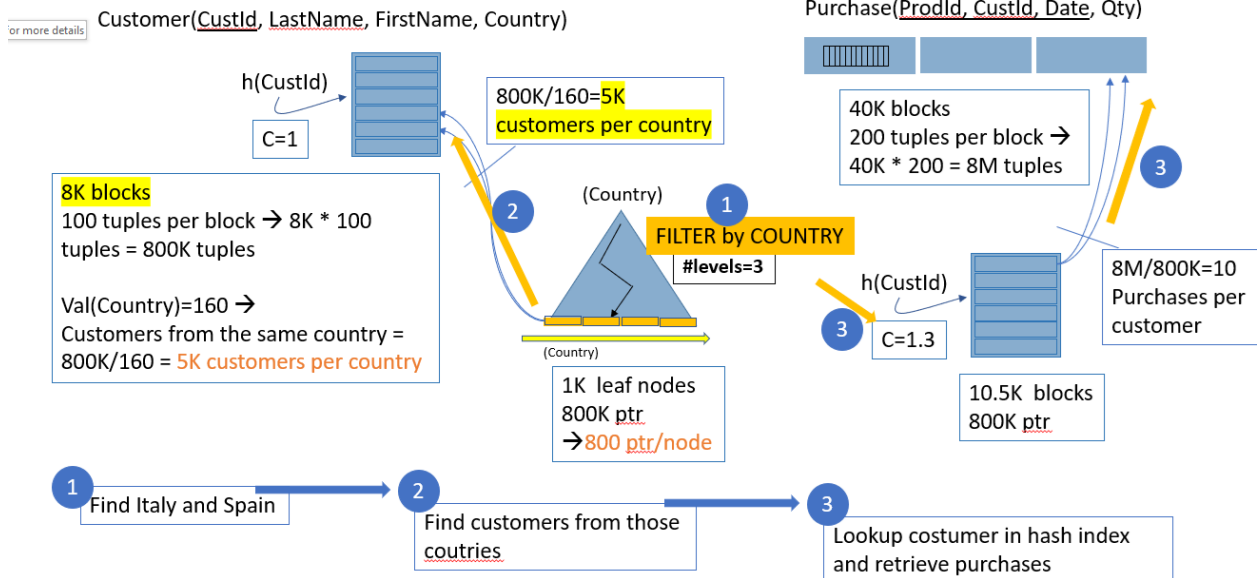
Plan5 = 8K + 10.5K + 2*(800K/160) * (8M/800K) = 8K + 10.5K + 10K * 10 = **118,5K** I/O accesses

Hash join + for each IT/ES customer follow the pointers



Best plan for scenario 2)

SCENARIO 3) Extend plan 4



Plan6 = 2 * (2 + (5K/800) + 5K) + 10K * (1.3 + 10) = **123K** I/O accesses

7 leaf nodes (*)
IT/ES int. levels and leaf nodes follow the pointers continue like in plan 4

(*) 2*5K > 8K (size of the hash table) For this query, the index cannot improve the costs

D. Theory (10 points, up to 2.5 points per question)

1. Briefly introduce R-trees (max 100 words).
2. Define the notion of MBB in an R-tree.
3. List and very briefly discuss the main 3 differences between R-trees and B⁺-trees.
4. Explain how the insertion of a new object in an R-tree works.

Solution.

See theory slides.